

Industry & Alumni Collaboration Portal

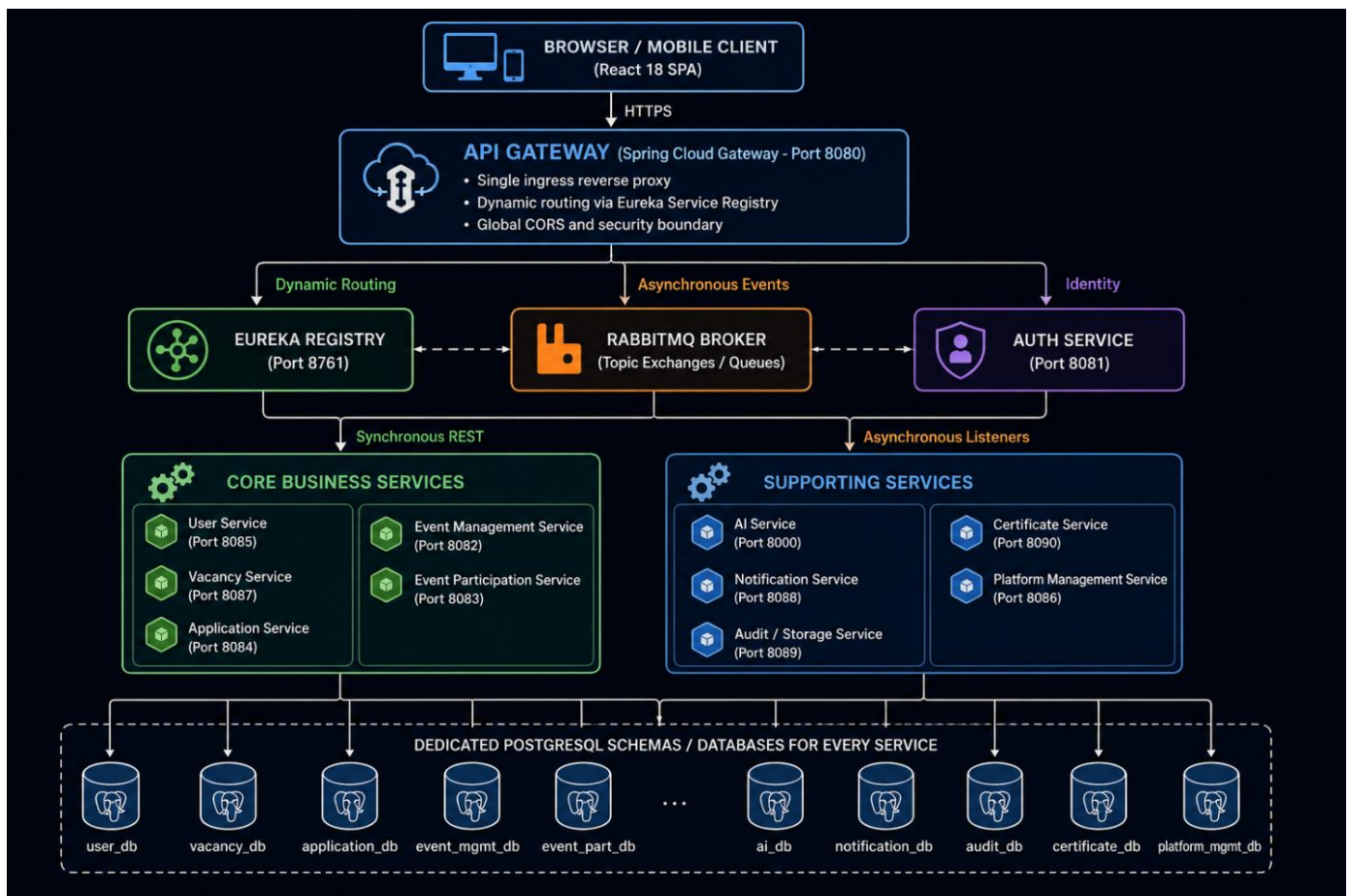
Live Cloud Production URL: <https://wonderful-wave-0320abf00.3.azurestaticapps.net/>

Core Stack: Spring Boot 3 (Java 17), Python 3.10 (FastAPI), React 18, Netflix Eureka, RabbitMQ, PostgreSQL

1. High-Level Architectural Overview

The **NSBM Industry & Alumni Collaboration Portal** architecture strictly decouples client interfaces from backing services through a reactive API Gateway and service discovery registry. Workflows are partitioned into **synchronous request-response cycles** (for real-time queries and authenticated UI actions) and **asynchronous event pipelines**.

2. Microservices Topology Diagram



3. End-to-End Operational Workflows

3.1 Student Registration & Profile Setup Flow

Step	Phase	Action & Inter-Service Collaboration
01	Registration Submission	Student completes the registration form (/register) selecting role STUDENT. Payload routes via API Gateway to Auth Service.
02	Credential Creation	Auth Service hashes password with BCrypt and creates record in auth_db.
03	Profile Shell Provisioning	User Service provisions an associated blank student profile shell linked to the newly generated userId.
04	Authentication	Student logs in via /login. Auth Service verifies credentials and issues a signed 24-hour JWT token.
05	Academic Portfolio Setup	Student populates degree program, faculty, department, GPA, expected graduation, and skills into user_db.

3.2 AI-Powered Resume Enhancement Flow

Step	Phase	Action & Inter-Service Collaboration
01	Resume Upload	Student uploads resume PDF in their profile modal. File routes to Audit & Storage Service (POST /api/v1/storage/upload), which returns a unique storageFileId.
02	Trigger Enhancement	Student clicks " Enhance Profile with AI ". User Service dispatches POST /api/v1/ai/resume/enhance-profile.
03	PDF Parsing & OCR	AI Service streams the PDF from Storage, extracts text via PyMuPDF, and prompts Google Gemini (gemini-3.1-flash-lite).
04	Structured Skill Extraction	Gemini returns validated technical proficiencies, frameworks, tools, and completed projects formatted as strict JSON.
05	Profile Enrichment	User Service updates the student's verified skills and project showcases, refreshing the frontend dashboard with AI-verified tags.

3.3 Industry Partner Onboarding & Institutional Accreditation

Step	Phase	Action & Inter-Service Collaboration
01	Partner Registration	Company representative registers on portal selecting INDUSTRY_PARTNER. Account status defaults to PENDING_VERIFICATION.
02	Document Submission	Partner uploads business registration certificates and corporate details via Audit & Storage Service.
03	Staff Audit Queue	University Placement Staff accesses Staff Admin Dashboard (/staff) and reviews pending partner applications.
04	Verification Approval	Staff verifies corporate legitimacy and clicks "Approve" . Platform Management updates partner status to VERIFIED.
05	Notification Dispatch	Notification Service dispatches an automated branded approval email. The partner is now unlocked to publish job vacancies.

3.4 Vacancy Creation, Flyer OCR & Gemini Processing Flow

Step	Phase	Action & Inter-Service Collaboration
01	Flyer Upload	Verified partner uploads job advertisement poster (PNG, JPG, or PDF) via Partner Portal (/partner/jobs).
02	Asset Persistence	Audit & Storage Service stores the binary graphic and returns storageFileId.
03	Event Emission	Vacancy Service creates a preliminary vacancy in status PENDING and publishes vacancy.flyer.process to RabbitMQ.
04	Vision OCR & Extraction	AI Service downloads graphic, runs Gemini Vision OCR, and extracts role title, seniority, workplace type, skills, and deadlines.
05	Anonymous Recruiter Correction	If recruiter provides a generic email (e.g., hrdcareers942@gmail.com), Gemini deduces the authentic enterprise brand name.
06	Curriculum Alignment	AI compares vacancy requirements against curriculum of all 4 NSBM faculties, scoring syllabus fit and noting anomalies.
07	Staff Review & Approval	University staff reviews AI recommendations on Staff Dashboard (/staff/vacancies) and clicks "Approve" .
08	Cohort Broadcasting	Vacancy transitions to APPROVED. Notification Service transmits alert emails to students in matching faculties.

3.5 Job Application & AI Semantic Match Scoring Flow

Step	Phase	Action & Inter-Service Collaboration
01	Vacancy Selection	Student navigates to job details and clicks "Apply" .
02	AI Cover Letter Synthesis	(Optional) Student clicks "Generate AI Cover Letter" . AI Service matches student profile against job requirements and synthesizes a tailored cover letter.
03	Application Submission	Student submits application. Application Service creates record in status SUBMITTED and emits application.submitted.match.
04	Semantic Embeddings Match	AI Service computes vector similarity embeddings between candidate profile/resume and job criteria.
05	Recruiter Ranking	AI returns a match score (0-100%), fit tier (HIGH_MATCH, GOOD_MATCH), and key candidate strengths.
06	Hiring Progression	Recruiter inspects AI-ranked candidate list and progresses status: SHORTLISTED → INTERVIEW → OFFERED.

3.6 Events, QR Attendance & Certificate Generation Flow

Step	Phase	Action & Inter-Service Collaboration
01	Event Creation	Staff or verified partner publishes career fair or hackathon with seat limits in Event Management Service.
02	Student Registration	Student registers. Event Participation Service reserves capacity and renders an encrypted QR code ticket in student ticket wallet.
03	Venue Scanner Check-in	Student presents ticket at entrance. Event organizer scans QR code using integrated portal mobile scanner.
04	Attendance Verification	Scanner marks attendance as ATTENDED and emits event.attendance.marked to RabbitMQ.
05	Tamper-Proof Certificate PDF	Certificate Service resolves attendee and event details, generating a branded PDF with NSBM seal and unique code (CERT-2026-XXXX).
06	Public Verification	Certificate embeds QR code pointing to public endpoint (/verify-certificate/{code}). Employers can verify validity without logging in.

4. Detailed Service-by-Service Technical Reference

Service 1: Frontend Client Application (frontend)

- **Technology:** React 18, Vite, React Router DOM, TailwindCSS, Lucide Icons, Axios
- **Ingress:** Client Root (/)
- **Database:** None (Client-side execution)
- **Core Responsibilities:**
 - Dynamic role-protected routing and persistent JWT token management via localStorage and Axios interceptors.
 - Real-time drag-and-drop resume upload and vacancy flyer OCR tracking.

Service 2: API Gateway (api-gateway)

- **Technology:** Spring Boot 3, Spring Cloud Gateway (Netty Reactive), Eureka Client
- **Port:** 8080 (and Azure Container App Ingress FQDN)
- **Database:** None (Stateless Reverse Proxy)
- **Core Responsibilities:**
 - Serves as single ingress boundary for all frontend and external requests.
 - Dynamic route resolution using Eureka registry (lb://<service-name>).
 - Enforces centralized Cross-Origin Resource Sharing (CORS) policies.
 - SSL termination, URL rewriting, and request header enrichment.
- **Route Mapping:**
 - /api/v1/auth/** → lb://auth-service
 - /api/v1/user-profiles/** → lb://user-service
 - /api/v1/vacancies/** → lb://vacancy-service
 - /api/v1/ai/** → lb://ai-service
 - /api/v1/applications/** → lb://application-service
 - /api/v1/events/** → lb://event-management-service
 - /api/v1/participations/** → lb://event-participation-service
 - /api/v1/certificates/** → lb://certificate-service
 - /api/v1/notifications/** → lb://notification-service
 - /api/v1/admin/** → lb://platform-management-service
 - /api/v1/storage/** → lb://audit-storage-service

Service 3: Discovery Server (discovery-server)

- **Technology:** Spring Cloud Netflix Eureka Server
- **Port:** 8761
- **Database:** In-Memory Dynamic Service Catalog
- **Core Responsibilities:**
 - Continuously registers microservice instances, container IP addresses, and operational health heartbeats.
 - Facilitates elastic horizontal scaling and zero-downtime rolling container updates on Azure Cloud.

Service 4: Auth Service (services/auth-service)

- **Technology:** Spring Boot 3, Spring Security 6, JWT, Spring Data JPA, PostgreSQL
- **Port:** 8081 | **Database:** auth_db
- **Base Endpoint:** /api/v1/auth/**
- **Core Responsibilities:**
 - Manages user registration, BCrypt password hashing, and role assignment (STUDENT, INDUSTRY_PARTNER, STAFF, SYSTEM_ADMIN).
 - Issues 24-hour cryptographically signed JWT access tokens containing user credentials and authorities.
 - Houses AuthDataInitializer, seeding default administrative, staff, corporate, and student test accounts upon startup.

Service 5: User Service (services/user-service)

- **Technology:** Spring Boot 3, Spring Data JPA, PostgreSQL
- **Port:** 8085 | **Database:** user_db
- **Base Endpoint:** /api/v1/user-profiles/**
- **Core Responsibilities:**
 - Manages rich academic student portfolios: GPA, degree program, faculty, department, and graduation year.
 - Maintains verified skill sets categorized by proficiency tiers with AI-verified flags.
 - Stores project showcase cards with GitHub repositories, live demonstration URLs, and technology badges.
 - Pre-seeded with complete authentic portfolios for all 10 standard test student profiles.

Service 6: Vacancy Service (services/vacancy-service)

- **Technology:** Spring Boot 3, Spring Data JPA, RabbitMQ Event Publisher, PostgreSQL
- **Port:** 8087 | **Database:** vacancy_db
- **Base Endpoint:** /api/v1/vacancies/**
- **Core Responsibilities:**
 - Manages corporate internship and job posting lifecycles: PENDING → APPROVED / REJECTED → CLOSED.
 - Ingests recruitment flyers and dispatches vacancy.flyer.process events to RabbitMQ.
 - Provides REST callback endpoints for AI Service to store parsed requirements and institutional alignment notes.
 - Delivers filtered search endpoints by faculty, required skills, workplace type, and location.

Service 7: AI Service (services/ai-service)

- **Technology:** Python 3.10, FastAPI, Google Gemini API, PyMuPDF, Sentence Transformers
- **Port:** 8000 | **Database:** ai_service_db (Internal SQLite / PostgreSQL)
- **Base Endpoint:** /api/v1/ai/**
- **Core Capabilities:**
 - **Flyer OCR & Gemini Extraction:** Employs Gemini Vision to transcribe flyer images and formats structured job attributes via gemini-3.1-flash-lite (with automatic failover across gemini-3.5-flash-lite and gemini-flash-lite-latest).
 - **Curriculum Alignment Checker:** Evaluates job requirements against all 4 NSBM university faculties, scoring curriculum suitability and flagging compliance issues.
 - **Anonymous Recruiter Branding:** Automatically translates generic contact emails into polished corporate identity names.
 - **Resume Profile Enhancer:** Extracts technical skills and project achievements from student CVs.
 - **Semantic Match Scoring:** Generates candidate-to-job match scores (0-100%) using vector similarity embeddings.
 - **AI Cover Letter Synthesizer:** Constructs bespoke cover letters matching applicant experience to job descriptions.

Service 8: Application Service (services/application-service)

- **Technology:** Spring Boot 3, Spring Data JPA, RabbitMQ Publisher, PostgreSQL
- **Port:** 8084 | **Database:** application_service_db
- **Base Endpoint:** /api/v1/applications/**
- **Core Responsibilities:**
 - Ingests student job applications with resume attachments and tailored cover letters.
 - Dispatches application.submitted.match events to RabbitMQ to initiate AI scoring.
 - Persists AI fit scores, fit tiers, and candidate strengths summaries.
 - Tracks corporate recruitment pipelines: SUBMITTED → SHORTLISTED → INTERVIEW → OFFERED → REJECTED.

Service 9: Event Management Service (services/event-management-service)

- **Technology:** Spring Boot 3, Spring Data JPA, RabbitMQ Publisher, PostgreSQL
- **Port:** 8082 | **Database:** event_management_db
- **Base Endpoint:** /api/v1/events/**
- **Core Responsibilities:**
 - Coordinates university career fairs, corporate hackathons, workshops, and guest lectures.
 - Enforces capacity limits, venue allocations, schedules, and target faculty restrictions.
 - Implements administrative approval queues for corporate partner-submitted events.

Service 10: Event Participation Service (services/event-participation-service)

- **Technology:** Spring Boot 3, Spring Data JPA, ZXing QR Engine, PostgreSQL
- **Port:** 8083 | **Database:** event_participation_db
- **Base Endpoint:** /api/v1/participations/**
- **Core Responsibilities:**
 - Manages real-time seat reservation quotas.
 - Generates cryptographically signed QR code ticket payloads for registered attendees.
 - Provides on-site organizer check-in verification endpoints.
 - Emits event.attendance.marked to RabbitMQ upon confirmed attendee scan.

Service 11: Certificate Service (services/certificate-service)

- **Technology:** Spring Boot 3, Spring Data JPA, iText / PDFBox Engine, PostgreSQL
- **Port:** 8090 | **Database:** certificate_db
- **Base Endpoint:** /api/v1/certificates/**
- **Core Responsibilities:**
 - Consumes event.attendance.marked to generate digital completion certificates.
 - Renders branded PDF certificates with the official NSBM crest, issue date, and attendee credentials.
 - Assigns verifiable serial codes (e.g., CERT-2026-XXXX) and embeds verification QR codes.
 - Provides public verification endpoints accessible without authentication.

Service 12: Notification Service (services/notification-service)

- **Technology:** Spring Boot 3, Spring Mail (JavaMailSender), RabbitMQ Listener, PostgreSQL
- **Port:** 8088 | **Database:** notification
- **Base Endpoint:** /api/v1/notifications/**
- **Core Responsibilities:**
 - Consumes notification queues across RabbitMQ.
 - Dispatches responsive, branded HTML emails for account verification, password resets, vacancy alerts, and certificates.
 - **Dynamic Domain Engine:** Auto-resolves URLs using the active cloud environment domain, preventing broken links.

Service 13: Platform Management Service (services/platform-management-service)

- **Technology:** Spring Boot 3, Spring Data JPA, PostgreSQL
- **Port:** 8086 | **Database:** platform_management_db
- **Base Endpoint:** /api/v1/admin/**
- **Core Responsibilities:**
 - Industry partner accreditation queue.
 - System-wide audit logging: tracks approvals, status alterations, and administrative operations.
 - Institutional dashboard compiling university placement.

Service 14: Audit & Storage Service (services/audit-storage-service)

- **Technology:** Spring Boot 3, Spring Data JPA, Storage Engine, PostgreSQL
- **Port:** 8089 | **Database:** audit_storage_db
- **Base Endpoint:** /api/v1/storage/**
- **Core Responsibilities:**
 - Centralized binary media repository (multipart/form-data) issuing GUID storageFileId keys.
 - Inline streaming support (?inline=true) for PDF document viewing and image rendering.
 - Houses resumes, recruitment flyers, business credentials, and event banners.
 - Tracks file metadata: MIME types, byte sizes, upload timestamps, and owner IDs.

Service 15: Common Security Library (services/common-security)

- **Technology:** Spring Security 6, JWT, Shared Maven Module
- **Port:** Embedded Library (N/A) | **Database:** None
- **Core Responsibilities:**
 - Standardized JWT authentication filter and token verification shared across all Spring microservices.
 - Establishes SecurityContextHolder with authenticated user IDs and role claims.
 - Provides method-level security annotations (@PreAuthorize("hasRole('...')")).
 - Standardizes HTTP error responses (401 Unauthorized, 403 Forbidden).

5. Pre-Registered Student Accounts for QA

Username	Account Role & Description	Password
student1	Student Account 1 (Computing - Software Engineering)	Student@1
student2	Student Account 2 (Computing - Data Science)	Student@2
student3	Student Account 3 (Computing - Cyber Security)	Student@3
student4	Student Account 4 (Business - International Business)	Student@4
student5	Student Account 5 (Business - Marketing & HR)	Student@5
student6	Student Account 6 (Business - Accounting & Finance)	Student@6
student7	Student Account 7 (Engineering - Mechatronics)	Student@7
student8	Student Account 8 (Engineering - Civil Engineering)	Student@8
student9	Student Account 9 (Science - Biomedical Science)	Student@9
student10	Student Account 10 (Science - Information Systems)	Student@10